

AIM : Implementing Substitution and Transposition Ciphers

1A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

CLI Version :

Code :

```
# Caesar Cipher
def caesar(text, key):
    ans = ""
    for ch in text.upper():
        if ch.isalpha():
            ans += chr((ord(ch)-65+key)%26+65)
        else:
            ans += ch
    return ans

# Playfair Cipher
def playfair(text, key, mode):
    alpha = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
    key = (key.upper()+alpha).replace("J","I")
    mat = []
    for ch in key:
        if ch not in mat and ch.isalpha():
            mat.append(ch)

    text = text.upper().replace(" ", "").replace("J","I")
    if len(text)%2:
        text += "X"

    ans = ""
    for i in range(0, len(text), 2):
        a, b = text[i], text[i+1]
        r1=c1=r2=c2=0

        for r in range(5):
            for c in range(5):
```

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Roll No : T085

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```
    if mat[r*5+c]==a:
        r1,c1=r,c
    if mat[r*5+c]==b:
        r2,c2=r,c

    if r1==r2:
        ans += mat[r1*5+(c1+mode)%5]
        ans += mat[r2*5+(c2+mode)%5]
    elif c1==c2:
        ans += mat[((r1+mode)%5)*5+c1]
        ans += mat[((r2+mode)%5)*5+c2]
    else:
        ans += mat[r1*5+c2]
        ans += mat[r2*5+c1]
    return ans

# Menu
while True:
    print("\n1.Caesar\n2.Playfair\n3.Exit")
    ch = input("Choice: ")

    if ch=="1":
        print("1.Encrypt\n2.Decrypt")
        op = input("Choice: ")
        msg = input("Message: ")
        key = int(input("Key: "))
        if op=="1":
            print("Encrypted:", caesar(msg,key))
        else:
            print("Decrypted:", caesar(msg,-key))

    elif ch=="2":
        print("1.Encrypt\n2.Decrypt")
        op = input("Choice: ")
        msg = input("Message: ")
        key = input("Keyword: ")
        if op=="1":
            print("Encrypted:", playfair(msg,key,1))
        else:
            print("Decrypted:", playfair(msg,key,-1))

    elif ch=="3":
```

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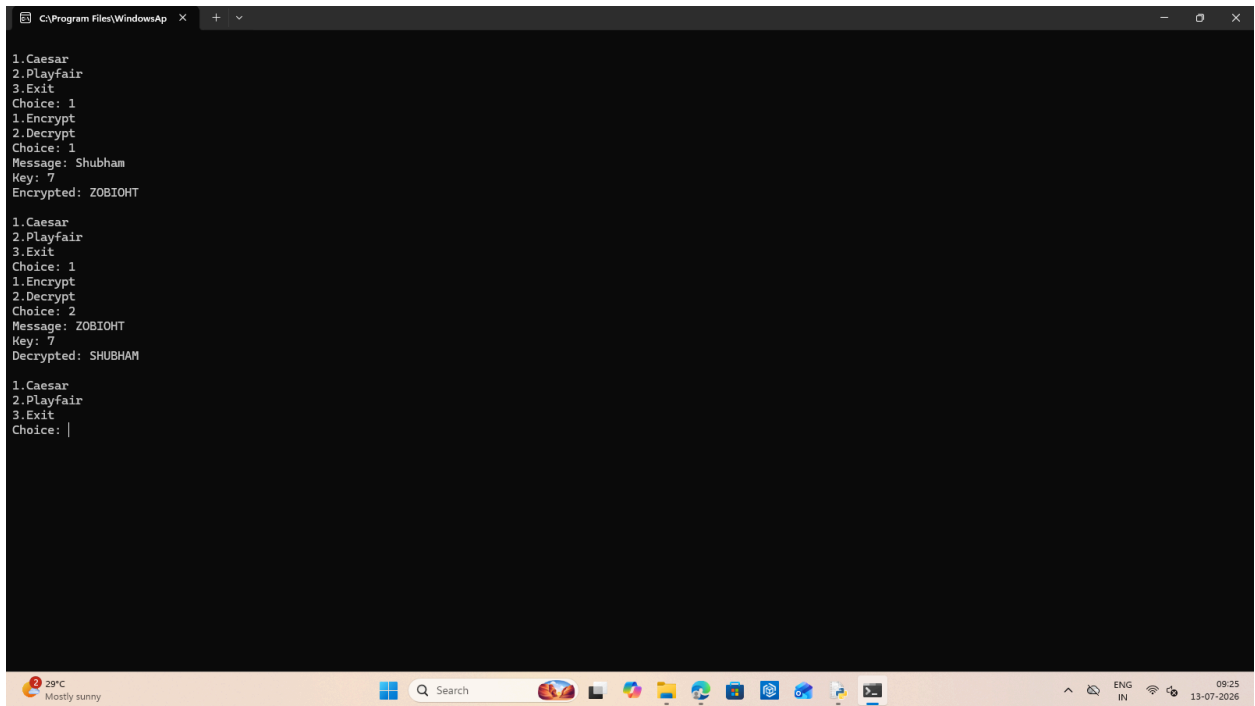
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```
print("Thank You")  
break
```

```
else:  
    print("Invalid Choice")
```

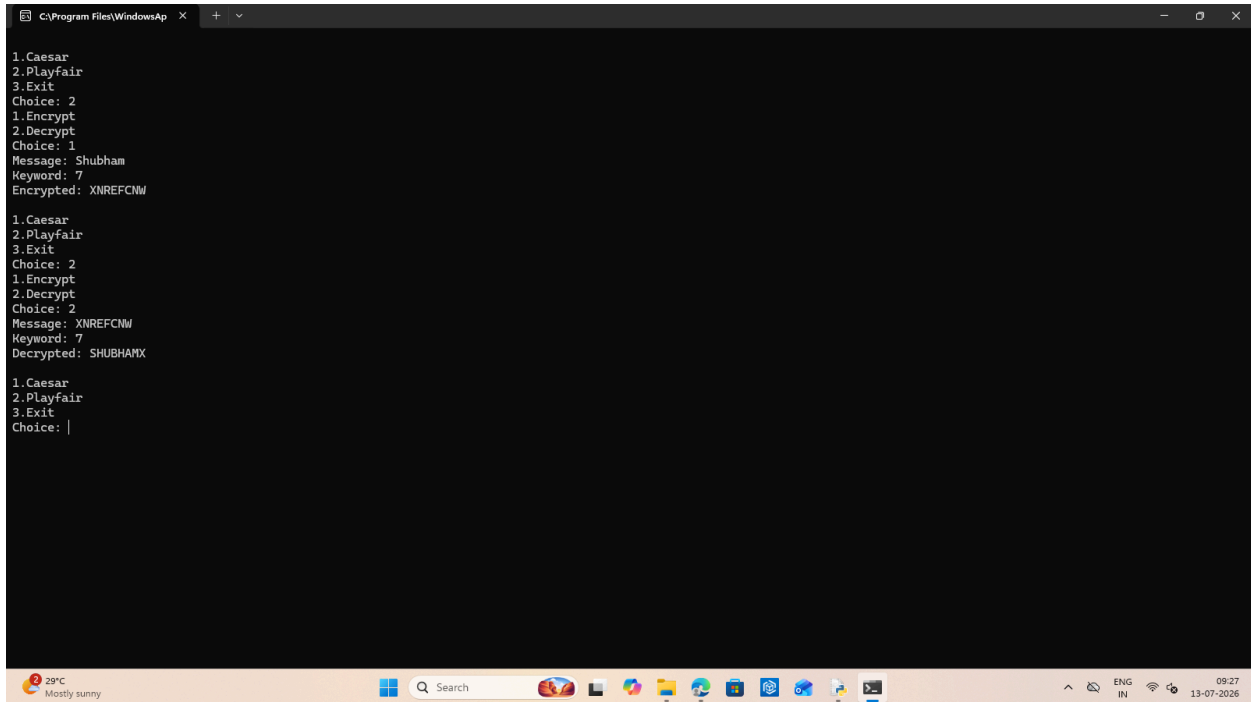
Output :



```
C:\Program Files\WindowsAp  x  +  v  
1.Caesar  
2.Playfair  
3.Exit  
Choice: 1  
1.Encrypt  
2.Decrypt  
Choice: 1  
Message: Shubham  
Key: 7  
Encrypted: ZOBIOHT  
  
1.Caesar  
2.Playfair  
3.Exit  
Choice: 1  
1.Encrypt  
2.Decrypt  
Choice: 2  
Message: ZOBIOHT  
Key: 7  
Decrypted: SHUBHAM  
  
1.Caesar  
2.Playfair  
3.Exit  
Choice: |
```

Name : Shubham Sanjay Karape
Roll No : T085

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```
C:\Program Files\WindowsAp  x  +  v
1.Caesar
2.Playfair
3.Exit
Choice: 2
1.Encrypt
2.Decrypt
Choice: 1
Message: Shubham
Keyword: 7
Encrypted: XNREFCNW

1.Caesar
2.Playfair
3.Exit
Choice: 2
1.Encrypt
2.Decrypt
Choice: 2
Message: XNREFCNW
Keyword: 7
Decrypted: SHUBHAMX

1.Caesar
2.Playfair
3.Exit
Choice: |
```

GUI Version :

Code :

```
from tkinter import *
```

```
def encrypt():
    msg = text.get()
    key = int(k.get())
    ans = ""
    for ch in msg.upper():
        if ch.isalpha():
            ans += chr((ord(ch)-65+key)%26+65)
        else:
            ans += ch
    result.config(text="Encrypted : "+ans)
```

```
def decrypt():
    msg = text.get()
    key = int(k.get())
```

Name : Shubham Sanjay Karape
Roll No : T085

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SUBJECT : CYBER SECURITY

```
ans = ""
for ch in msg.upper():
    if ch.isalpha():
        ans += chr((ord(ch)-65-key)%26+65)
    else:
        ans += ch
result.config(text="Decrypted : "+ans)

root = Tk()
root.title("Caesar Cipher")
root.state("zoomed")
root.configure(bg="black")

Label(root,text="CAESAR CIPHER",font=("Arial",24,"bold"),
      bg="black",fg="cyan").pack(pady=20)

Label(root,text="Message",bg="black",fg="white").pack()
text = Entry(root,width=40,font=15)
text.pack()

Label(root,text="Key",bg="black",fg="white").pack()
k = Entry(root,width=10,font=15)
k.pack()

Button(root,text="Encrypt",bg="cyan",fg="black",
       command=encrypt,width=15).pack(pady=10)

Button(root,text="Decrypt",bg="cyan",fg="black",
       command=decrypt,width=15).pack()

result = Label(root,text="",font=("Arial",16),
              bg="black",fg="lime")
result.pack(pady=20)

Button(root,text="Exit",command=root.destroy,
      bg="red",fg="white",width=10).pack()

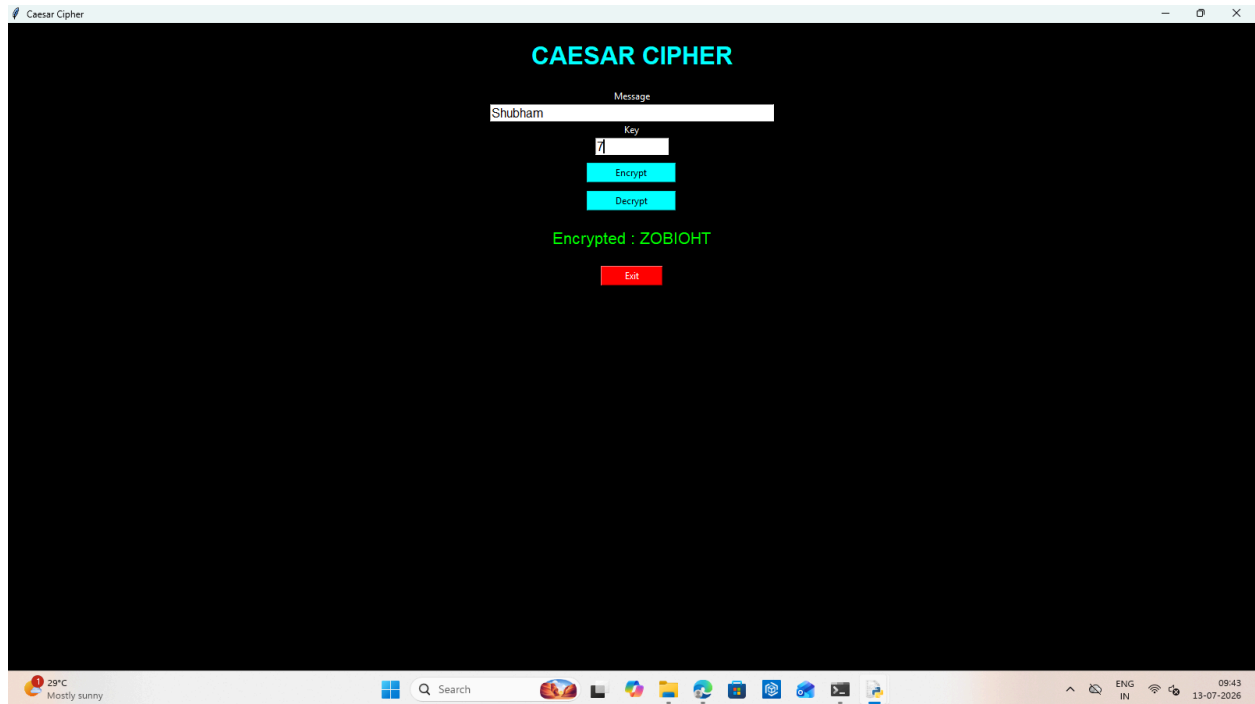
root.mainloop()
```

Name : Shubham Sanjay Karape

Roll No : T085

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Output :



1B Design and implement algorithms to encrypt and decrypt messages using transposition techniques

CLI Version :

Code :

```
# Rail Fence Cipher
def rail_encrypt(text):
    odd = text[::2]
    even = text[1::2]
    return odd + even
```

```
def rail_decrypt(text):
    mid = (len(text)+1)//2
    odd = text[:mid]
    even = text[mid:]
    result = ""
```

Name : Shubham Sanjay Karape
Roll No : T085

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SUBJECT : CYBER SECURITY

```
for i in range(len(even)):
    result += odd[i] + even[i]
if len(odd) > len(even):
    result += odd[-1]
return result
```

Columnar Transposition Cipher

```
def column_encrypt(text, key):
    while len(text) % key != 0:
        text += "X"
    result = ""
    for i in range(key):
        result += text[i::key]
    return result
```

```
def column_decrypt(text, key):
    rows = len(text) // key
    cols = []
    for i in range(key):
        cols.append(text[i*rows:(i+1)*rows])
    result = ""
    for r in range(rows):
        for c in range(key):
            result += cols[c][r]
    return result
```

Main Menu

```
while True:
    print("\n===== Transposition Cipher =====")
    print("1. Rail Fence")
    print("2. Columnar")
    print("3. Exit")

    ch = input("Enter Choice: ")

    if ch == "1":
        print("\n1. Encrypt")
        print("2. Decrypt")
        op = input("Enter Choice: ")
```

Name : Shubham Sanjay Karape

Roll No : T085

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SUBJECT : CYBER SECURITY

```
msg = input("Enter Message: ")

if op == "1":
    print("Encrypted:", rail_encrypt(msg))
elif op == "2":
    print("Decrypted:", rail_decrypt(msg))

elif ch == "2":
    print("\n1. Encrypt")
    print("2. Decrypt")
    op = input("Enter Choice: ")

    msg = input("Enter Message: ")
    key = int(input("Enter Key: "))

    if op == "1":
        print("Encrypted:", column_encrypt(msg, key))
    elif op == "2":
        print("Decrypted:", column_decrypt(msg, key))

elif ch == "3":
    print("Program Ended")
    break

else:
    print("Invalid Choice")
```


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Output :

```
C:\Program Files\WindowsAp  x  +  v

===== Transposition Cipher =====
1. Rail Fence
2. Columnar
3. Exit
Enter Choice: 1

1. Encrypt
2. Decrypt
Enter Choice: 1
Enter Message: Shubham
Encrypted: Suhmhba

===== Transposition Cipher =====
1. Rail Fence
2. Columnar
3. Exit
Enter Choice: 1

1. Encrypt
2. Decrypt
Enter Choice: 2
Enter Message: Suhmhba
Decrypted: mShubha

===== Transposition Cipher =====
1. Rail Fence
2. Columnar
3. Exit
Enter Choice: |
```

```
C:\Program Files\WindowsAp  x  +  v

1. Rail Fence
2. Columnar
3. Exit
Enter Choice: 2

1. Encrypt
2. Decrypt
Enter Choice: 2
Enter Message: shubham
Enter Key: 1
Decrypted: shubham

===== Transposition Cipher =====
1. Rail Fence
2. Columnar
3. Exit
Enter Choice: 2

1. Encrypt
2. Decrypt
Enter Choice: 1
Enter Message: Shubham
Enter Key: 31
Encrypted: ShubhamXXXXXXXXXXXXXXXXXXXXXXXXXXXX

===== Transposition Cipher =====
1. Rail Fence
2. Columnar
3. Exit
Enter Choice: |
```

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Roll No : T085

GUI Version :

Code :

```
from tkinter import *
from tkinter import messagebox

# Rail Fence
def rail_encrypt(t):
    return t[::2] + t[1::2]

def rail_decrypt(t):
    m = (len(t)+1)//2
    a, b = t[:m], t[m:]
    r = ""
    for i in range(len(b)):
        r += a[i] + b[i]
    if len(a) > len(b):
        r += a[-1]
    return r

# Columnar
def column_encrypt(t, k):
    while len(t) % k != 0:
        t += "X"
    return "".join(t[i::k] for i in range(k))

def column_decrypt(t, k):
    rows = len(t)//k
    col = [t[i*rows:(i+1)*rows] for i in range(k)]
    return "".join(col[c][r] for r in range(rows) for c in range(k))

# Process
def run():
    msg = e1.get()

    if cipher.get() == "Rail":
        ans = rail_encrypt(msg) if op.get() == "Encrypt" else rail_decrypt(msg)
    else:
        try:
            key = int(e2.get())
```

Name : Shubham Sanjay Karape

Roll No : T085

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SUBJECT : CYBER SECURITY

```
        ans = column_encrypt(msg, key) if op.get() == "Encrypt" else
column_decrypt(msg, key)
    except:
        messagebox.showerror("Error", "Enter Key")
        return

    out.config(text=ans)

# GUI
root = Tk()
root.title("Transposition Cipher")
root.geometry("400x350")

cipher = StringVar(value="Rail")
op = StringVar(value="Encrypt")

Label(root, text="Cipher").pack()
OptionMenu(root, cipher, "Rail", "Columnar").pack()

Label(root, text="Operation").pack()
OptionMenu(root, op, "Encrypt", "Decrypt").pack()

Label(root, text="Message").pack()
e1 = Entry(root, width=30)
e1.pack()

Label(root, text="Key (Columnar)").pack()
e2 = Entry(root, width=10)
e2.pack()

Button(root, text="Run", command=run).pack(pady=10)

out = Label(root, text="", font=("Arial", 12))
out.pack()

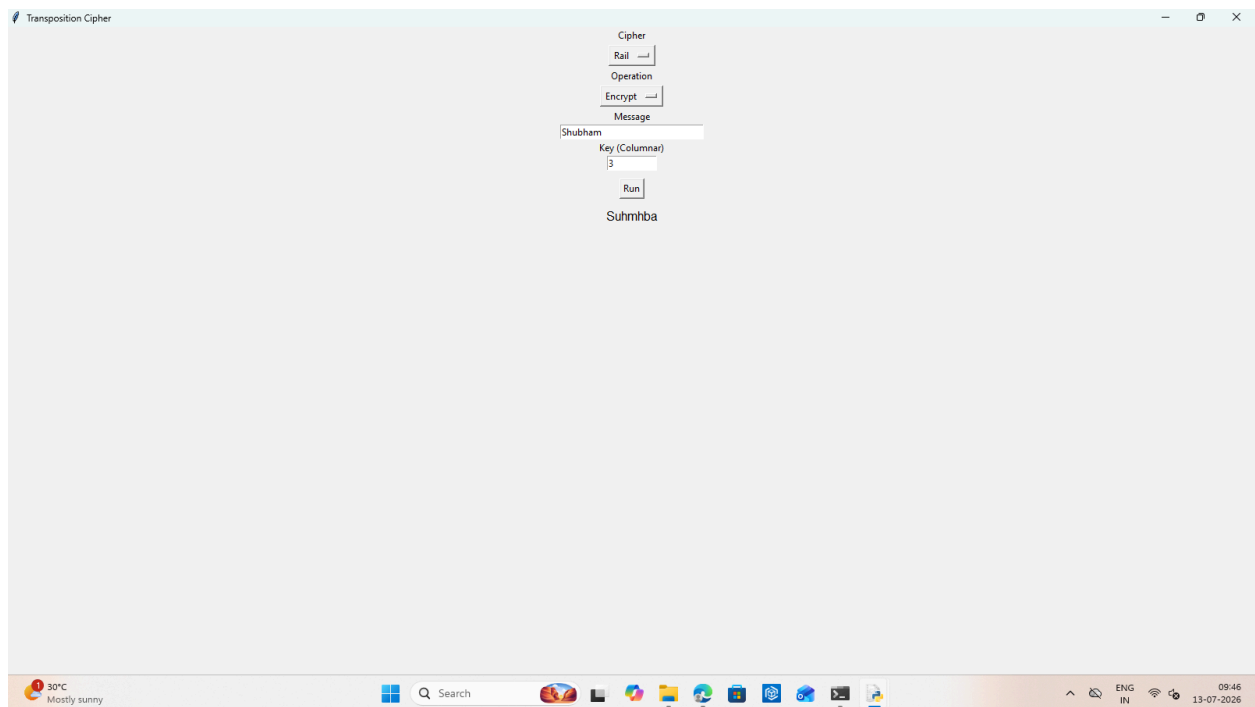
root.mainloop()
```

Name : Shubham Sanjay Karape

Roll No : T085

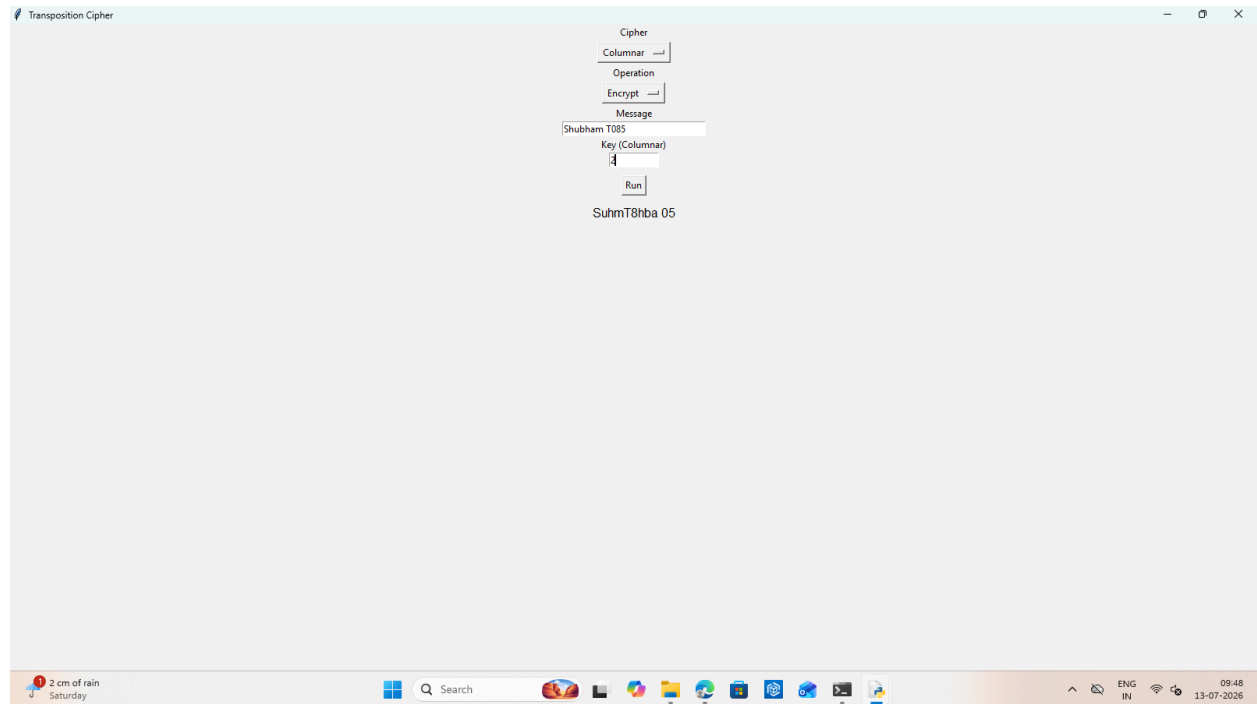
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SUBJECT : CYBER SECURITY

Output :



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Roll No : T085

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Roll No : T085